b230e1.c

b230e1.c > main()

```c
#include <stdio.h>
#include <string.h>

#define MAX 10

char stack[MAX][50];
int top = -1;

void push(char alert[]) {
    if (top == MAX - 1) {
        printf("Stack Overflow\n");
    } else {
        top++;
        strcpy(stack[top], alert);
        printf("PUSHED: %s\n", alert);
    }
}

void popAll() {
    if (top == -1) {
        printf("No active alerts\n");
    } else {
        printf("\nSoil moisture normal. Removing active alerts:\n");
        while (top != -1) {
            printf("POPPED: %s\n", stack[top]);
            top--;
        }
    }
}

void displayStack() {
    if (top == -1) {
        printf("Stack is Empty\n");
    } else {
        printf("\nCurrent Active Alerts:\n");
        for (int i = top; i >= 0; i--) {
            printf("%s\n", stack[i]);
```

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```c
31  void displayStack() {
34      } else {
35          printf("\nCurrent Active Alerts:\n");
36          for (int i = top; i >= 0; i--) {
37              printf("%s\n", stack[i]);
38          }
39          printf("\n");
40      }
41  }
42
43  void checkMoisture(float moisture) {
44      printf("\nSoil Moisture Level = %.1f%%\n", moisture);
45
46      if (moisture <= 15.0) {
47          push("Critical Alert: High Irrigation Needed");
48      } else if (moisture <= 25.0) {
49          push("Warning: Moderate Water Deficit");
50      } else if (moisture <= 40.0) {
51          push("Advisory: Low Moisture");
52      } else {
53          popAll();
54      }
55  }
56
57  int main() {
58      checkMoisture(45.0);
59      displayStack();
60
61      checkMoisture(35.0);
62      displayStack();
63
64      checkMoisture(22.0);
65      displayStack();
66
67      checkMoisture(12.0);
68      displayStack();
```

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```
43  void checkMoisture(float moisture) {
55  }
56
57  int main() {
58      checkMoisture(45.0);
59      displayStack();
60
61      checkMoisture(35.0);
62      displayStack();
63
64      checkMoisture(22.0);
65      displayStack();
66
67      checkMoisture(12.0);
68      displayStack();
69
70      checkMoisture(50.0);
71      displayStack();
72
73      return 0;
74  }
```

```
Soil Moisture Level = 45.0%
No active alerts
Stack is Empty

Soil Moisture Level = 35.0%
PUSHED: Advisory: Low Moisture

Current Active Alerts:
Advisory: Low Moisture


Soil Moisture Level = 22.0%
PUSHED: Warning: Moderate Water Deficit

Current Active Alerts:
Warning: Moderate Water Deficit
Advisory: Low Moisture


Soil Moisture Level = 12.0%
PUSHED: Critical Alert: High Irrigation Needed

Current Active Alerts:
Critical Alert: High Irrigation Needed
Warning: Moderate Water Deficit
Advisory: Low Moisture


Soil Moisture Level = 50.0%

Soil moisture normal. Removing active alerts:
POPPED: Critical Alert: High Irrigation Needed
POPPED: Warning: Moderate Water Deficit
POPPED: Advisory: Low Moisture
Stack is Empty
```